

Deterministic Optimization

Large-Scale Optimization

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Example for Column Generation

Large-Scale Optimization

Learning Objectives for this module

- Apply column generation algorithm to numerically solve a cutting stock problem

Column Generation Algorithm

We want to solve a large-scale LP with many variables

1. Pick a subset I of columns $\{A_j\}$ and variables x_j for $j \in I \subseteq \{1, 2, \dots, N\}$
2. Solve (RMP) with a subset of columns I

$$\begin{aligned} (MP) \quad & \min \sum_{j=1}^N x_j \\ & \text{s.t.} \quad \sum_{j=1}^N A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

$$\begin{aligned} (RMP) \quad & \min \sum_{j \in I} x_j \\ & \text{s.t.} \quad \sum_{j \in I} A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j \in I. \end{aligned}$$

Solving (RMP) is not challenging (and it shouldn't be) because of the small number of columns.

Column Generation Algorithm (cont.)

3. Pricing problem

$$\hat{Z} = \max_{j=1, \dots, N} \hat{\mathbf{y}}^T \mathbf{A}_j = \max \quad \hat{\mathbf{y}}^T \mathbf{A}$$

s.t. $\mathbf{A}$ is a feasible column for the cutting stock problem

$$= \max \quad \sum_{i=1}^m \hat{y}_i a_i \tag{1}$$

$$\text{s.t.} \quad \sum_{i=1}^m w_i a_i \leq W \tag{2}$$

$$a_i \geq 0, \text{ integer.} \tag{3}$$

- If $1 - \hat{Z} \geq 0$, all reduced costs are nonnegative. Terminate.
- If $1 - \hat{Z} < 0$, add the generated column $[a_1; a_2; a_3]$ to (RMP). Go to Step 1.

Numerical Example

- Large roll width $W = 100$
- Demand: $w_1 = 25$, $b_1 = 150$; $w_2 = 35$, $b_2 = 200$; $w_3 = 45$, $b_3 = 300$
- Patterns

Item	Patterns													Demand
w_i	1	2	3	4	5	6	7	8	9	10	11		N	b_i
25	4	3	2	2	2	1	1	1	1	0	0		a_1	150
35	0	0	0	1	0	2	0	1	0	1	0		a_2	200
45	0	0	0	0	1	0	1	0	0	0	1		a_3	300

Iteration 1: Solve RMP

- Initial set of patterns and RMP

$$\begin{aligned} \min \quad & x_9 + x_{10} + x_{11} \\ \text{s.t.} \quad & \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_9 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} x_{10} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x_{11} = \begin{bmatrix} 150 \\ 200 \\ 300 \end{bmatrix} \\ & x_9, x_{10}, x_{11} \geq 0. \end{aligned}$$

- Three demands (linear equality constraints) → three patterns (columns)
- Optimal solution: $\bar{x}_9 = 150, \bar{x}_{10} = 200, \bar{x}_{11} = 300$

- Optimal basis: $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad c_B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \hat{y}^T = c_B^T B^{-1} = [1, 1, 1]$

Iteration 1: Pricing Problem

- Form and solve knapsack problem

$$\begin{aligned}\hat{Z} = \max \quad & a_1 + a_2 + a_3 \\ \text{s.t.} \quad & 25a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integers}\end{aligned}$$

The optimal solution is $a_1 = 4, a_2 = 0, a_3 = 0$, which corresponds to $\mathbf{A}_1$. The objective value $Z = 4$. Therefore, the minimum reduced cost is $1 - \hat{Z} = -3 < 0$, which means the current solution $\hat{\mathbf{x}}$ is not optimal. Column $\mathbf{A}_1$ enters the list I .

Iteration 2: RMP

- Set of columns $I = \{1,9,10,11\}$
- Solve RMP

$$\begin{aligned} \min \quad & x_1 + x_9 + x_{10} + x_{11} \\ \text{s.t.} \quad & \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_9 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} x_{10} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x_{11} = \begin{bmatrix} 150 \\ 200 \\ 300 \end{bmatrix} \\ & x_1, x_9, x_{10}, x_{11} \geq 0. \end{aligned}$$

- Optimal solution: $\bar{x}_1 = 37.5, \bar{x}_9 = 0, \bar{x}_{10} = 200, \bar{x}_{11} = 300$
- Optimal basis:

$$B = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad c_B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \hat{y}^T = c_B^T B^{-1} = [1/4, 1, 1]$$

Iteration 2: Pricing Problem

- Compute minimum reduced cost

$\min_{j=1,\dots,N} \{1 - \hat{\mathbf{y}}^T \mathbf{A}_j\}$. Equivalently, solve the knapsack problem:

$$\begin{aligned} \hat{Z} = \max \quad & \frac{1}{4}a_1 + a_2 + a_3 \\ \text{s.t.} \quad & 25a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integers} \end{aligned}$$

The optimal solution is $a_1 = 1, a_2 = 2, a_3 = 0$, which corresponds to $\mathbf{A}_6$. The objective value $Z = 9/4$. Therefore, the minimum reduced cost is $1 - \hat{Z} = -5/4 < 0$, which means the current solution $\hat{\mathbf{x}}$ is not optimal. Column $\mathbf{A}_6$ enters the list I .

Iteration 3: RMP

- Set of columns $I = \{1,6,9,10,11\}$
- Solve RMP

$$\begin{aligned} \min \quad & x_1 + x_6 + x_9 + x_{10} + x_{11} \\ \text{s.t.} \quad & \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} x_6 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_9 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} x_{10} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x_{11} = \begin{bmatrix} 150 \\ 200 \\ 300 \end{bmatrix} \\ & x_1, x_6, x_9, x_{10}, x_{11} \geq 0. \end{aligned}$$

- Optimal solution: $\bar{x}_1 = 12.5, \bar{x}_6 = 100, \bar{x}_9 = 0, \bar{x}_{10} = 0, \bar{x}_{11} = 300$
- Optimal basis:

$$B = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad c_B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \hat{y} = c_B^\top B^{-1} = [1/4 \quad 3/8 \quad 1]$$

Iteration 3: Pricing Problem

- Compute minimum reduced cost

$\min_{j=1,\dots,N} \{1 - \hat{\mathbf{y}}^T \mathbf{A}_j\}$. Equivalently, solve the knapsack problem:

$$\begin{aligned} \hat{Z} = \max \quad & \frac{1}{4}a_1 + \frac{3}{8}a_2 + a_3 \\ \text{s.t.} \quad & 25a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integers} \end{aligned}$$

The optimal solution is $a_1 = 0, a_2 = 0, a_3 = 2$, which corresponds to $\mathbf{A}_{14}$. The objective value $Z = 2$. Therefore, the minimum reduced cost is $1 - \hat{Z} = -1 < 0$, which means the current solution $\hat{\mathbf{x}}$ is not optimal. Column $\mathbf{A}_{14}$ enters the list I .

Iteration 4: RMP

- Set of columns $I = \{1,6,9,10,11,14\}$
- Solve RMP

$$\begin{aligned} \min \quad & x_1 + x_6 + x_9 + x_{10} + x_{11} + x_{14} \\ \text{s.t.} \quad & \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} x_6 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_9 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} x_{10} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} x_{11} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} x_{14} = \begin{bmatrix} 150 \\ 200 \\ 300 \end{bmatrix} \\ & x_1, x_6, x_9, x_{10}, x_{11}, x_{14} \geq 0. \end{aligned}$$

- Optimal solution: $\bar{x}_1 = 12.5, \bar{x}_6 = 100, \bar{x}_9 = \bar{x}_{10} = \bar{x}_{11} = 0, \bar{x}_{14} = 150$
- Optimal basis:

$$B = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad c_B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \hat{y} = c_B^\top B^{-1} = [1/4 \quad 3/8 \quad 1/2]$$

Iteration 4: Pricing Problem

- Compute minimum reduced cost

$\min_{j=1,\dots,N} \{1 - \hat{\mathbf{y}}^T \mathbf{A}_j\}$. Equivalently, solve the knapsack problem:

$$\begin{aligned} \hat{Z} = \max \quad & \frac{1}{4}a_1 + \frac{3}{8}a_2 + \frac{1}{2}a_3 \\ \text{s.t.} \quad & 25a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integers} \end{aligned}$$

The optimal solution is $a_1 = 2, a_2 = 0, a_3 = 1$. The objective value $Z = 1$. Therefore, the minimum reduced cost is $1 - \hat{Z} = 0$, which means the current solution $\hat{\mathbf{x}}$ is optimal!

Summary

- We learned:
 - The detailed steps of applying column generation to the cutting stock problem